MEASURING MACROECONOMIC AND FINANCIAL UNCERTAINTY: IMPACT FOR THE MEXICAN ECONOMIC GROWTH EXPECTATION

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Motivation

UNCERTAINTY IN MACROECONOMICS

Greenspan (2003): “Uncertainty is not just an important feature of the monetary policy landscape; it is the defining characteristic of that landscape”

Some concepts of uncertainty in the literature:

- **EPU** - (Baker, Bloom and Davis) How uncertainty is the Policy Environment that households and firms face.
- **VIX** - How much financial markets expect asset prices to fluctuate.
- **JLN** - (Jurado, Ludvigson and Ng) How unpredictable in macroeconomy.
- **LMN** - (Ludvigson, Ma and Ng) What are the roles of **macroeconomic** and **financial** uncertainty.

WHY IS THIS IMPORTANT?

Effects of heightened uncertainty

- More dispersion and negative skewed future growth distribution of economic activity
- Non-linear and asymmetric responses to uncertainty
- Higher asset prices volatility

Remark: Monetary policy makers should conducts a balance of risks weighing downside risks —shocks to production, short and long term unemployment— and upside risks—inflation, to determine the entire probability distribution of expected economic growth.

In this context Adrian, Boyarchenko and Gianone (2019), propose a methodology to determine *what is the worst GDP outcome that we should expect with given probability?*

LITERATURE REVIEW – FOR THE MEXICAN ECONOMY

- Salgado and Trujillo (2025). How macroeconomic uncertainty, using the Financial Conditions Index (Carrillo and Garcia, 2021), affects growth expectations.
- Catalán (2019) How uncertainty shocks from EPU affect prices (interest rates, wages) and, consequently, economic activity (investment and employment).
- Cebreros (2000) How trade policy uncertainty affects FDI.

RESEARCH QUESTION

- **What role does macroeconomic and financial uncertainty play in the determination of growth expectation for the Mexican economy?**
- **Contribution:** disentangle the MU from FU and apply the GaR methodology.

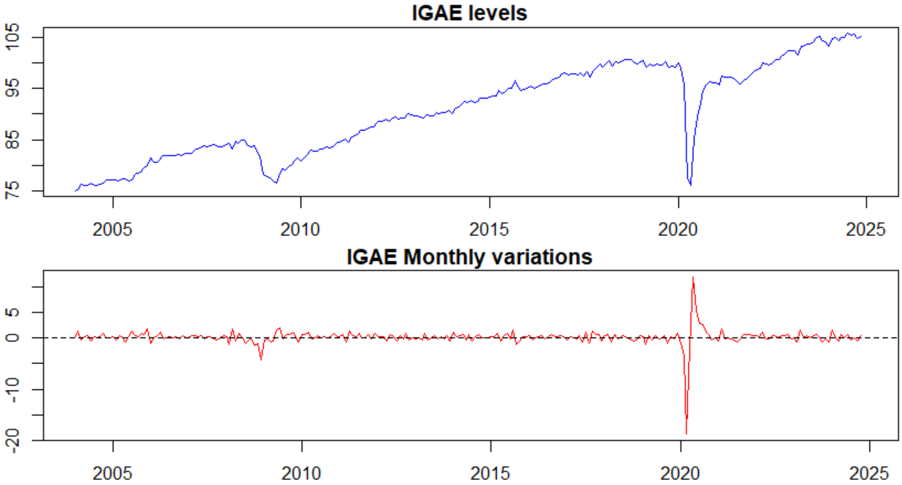
Methodology

METHODOLOGY

1. Estimate common trends to the Mexican economy, following the work by Corona et al (2021). They develop a panel of 26 variables that are used to nowcast Mexican monthly economic activity indicator (IGAE).
2. Compute *macro* uncertainty, MU, and *financial* uncertainty, FU, indicators, following Ludvignson, Ma and Ng (2021), as the unforeseeable component of economic variables after extracting all common predictable variation in the DFM.
3. Estimate a SVAR to identify causal direction between MU, FU and economic activity.
4. Estimating the entire conditional distribution of growth for evaluating uncertainty's impact in the Mexican economy (Adrian, Boyarchenko and Gianone, 2019).

Data

MEXICAN ECONOMIC ACTIVITY INDICATOR



UNIT ROOT TESTS

Series	Test	Stat	CV 1%	CV 5%	CV 10%
ADF — H_0: unit root; reject if stat < CV					
Levels	ADF	−4.792	−3.980	−3.420	−3.130
Monthly var.	ADF	−13.030	−3.460	−2.880	−2.570
PP — H_0: unit root; reject if stat < CV					
Levels	PP	−3.761	−3.998	−3.429	−3.138
Monthly var.	PP	−11.934	−3.458	−2.873	−2.573
KPSS — H_0: stationary; reject if stat > CV					
Levels	KPSS	0.203	0.216	0.176	0.146
Monthly var.	KPSS	0.022	0.739	0.574	0.463

For levels evidence is mixed — ADF/PP reject the unit root at 5% but KPSS also rejects stationarity at 5%.

Monthly variation is $I(0)$: ADF and PP reject the unit root, and KPSS does not reject stationarity at any conventional level.

This validates using IGAE monthly growth rate in the VAR.

STRUCTURAL BREAKS AT MEXICAN RECESSION DATES

Series	Date	F	p-val
IGAE levels			
Levels	2009/05	243.86	0.000
Levels	2020/05	115.32	0.000
IGAE monthly variation			
Monthly var.	2009/05	0.271	0.603
Monthly var.	2020/05	1.657	0.199

Chow test, H_0: no structural break.

Dates: CEEY recession chronology.

Structural breaks are present in the *level* of IGAE but **not** in its monthly growth rate.

- The 2009 and 2020 recessions produced **permanent level shifts** in economic activity.
- Monthly variation does not reflect this shifts is **structurally stable** across both episodes ($p > 0.10$).
- First-differencing simultaneously removes the unit root and the structural breaks.

ENDOGENOUS STRUCTURAL BREAKS — IGAE

Bai-Perron (BIC-optimal partition)

	Levels	Monthly var.
Optimal breaks (m)	6	0
BIC	1234	963.7

Levels — break dates ($m = 6$):

Date	Episode
2006/01	Post-NAFTA adjustment
2011/05	Euro-area sovereign crisis
2014/03	Oil price decline
2016/08	US electoral uncertainty
2020/02	COVID-19 recession
2022/03	Post-pandemic inflation surge

Monthly variation — no breaks detected.

Zivot-Andrews

Series	Stat	CV 1%	Break
Levels	−5.697	−5.57	2020/02
Monthly var.	−12.797	−5.34	2020/04

Both reject unit root at 1%.

- Bai-Perron identifies **six level shifts** in IGAE, concentrated around macro events.
- Monthly growth rate shows **no structural breaks** — the BIC selects $m = 0$.
- Zivot-Andrews confirms break at ($-5.70 < -5.57$) and rejects the unit root for monthly variation ($-12.80 \ll -5.34$).

DATA FOR DFM

The panel time series consists of 26 variables with the following characteristics:

- Monthly frequency,
- Starting from January 2004, with some available through January 2025.
- All variables have been obtained in seasonally adjusted form; if not, they have been adjusted using the X-13ARIMA-SEATS method. This seasonal adjustment relies on the automatic modeling procedure implemented in the seas function from R's seasonal package.
- Optimal transformation is applied first-differences, monthly variation, yearly variation such that it maximize the correlation with IGAE.

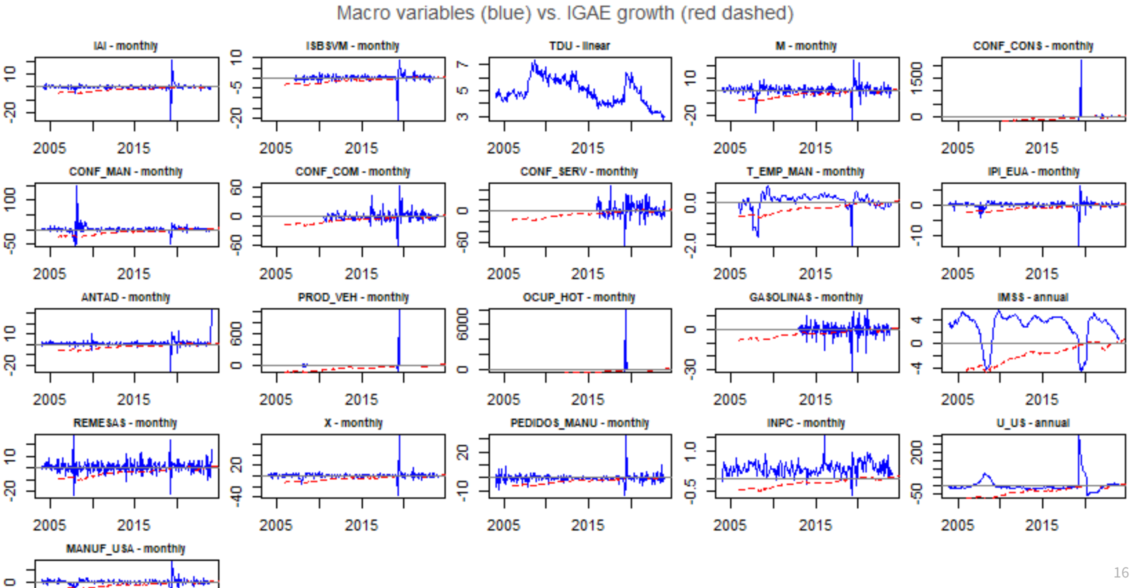
DATA: VARIABLE PANEL

Variable	Description	Trans.
Macro variables		
IMAI	Industrial production	monthly
Sales	Retail sales	monthly
U	Unemployment	monthly
M	Imports	monthly
Conf Constr	Construction confidence	monthly
Conf Manuf	Manufacturing confidence	monthly
Conf Trade	Trade confidence	monthly
Conf Serv	Services confidence	monthly
Trend L Manuf	Manuf. employment trend	monthly
Industrial USA	U.S. industrial production	monthly
ANTAD	ANTAD retail sales	monthly
Automotive	Vehicle production	monthly
Hotel	Hotel occupancy	linear
Gas	Gasoline sales	monthly

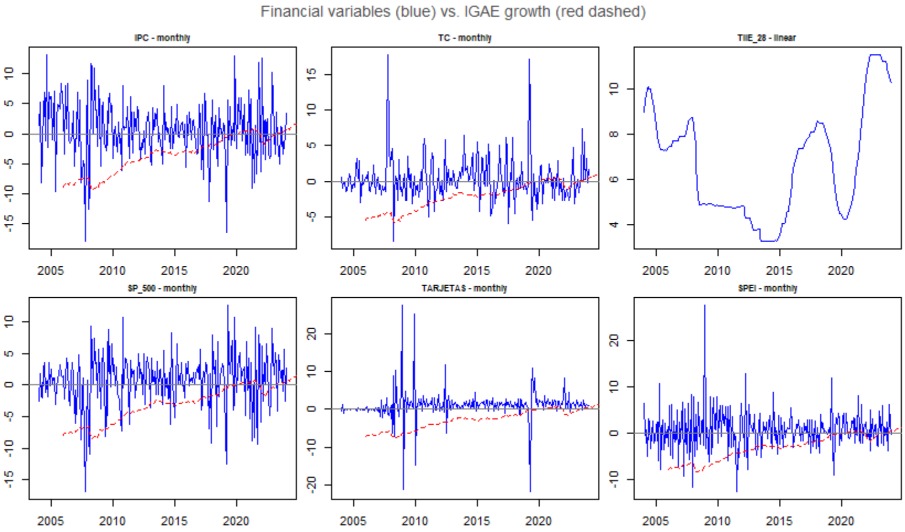
Variable	Description	Trans.
Macro variables (cont.)		
IMSS	IMSS formal employment	monthly
Remittances	Remittances inflows	monthly
X	Exports	monthly
PMI	Manufacturing orders	monthly
U USA	U.S. unemployment rate	monthly
Manuf USA	U.S. manufacturing activity	monthly
PIBO	Timely GDP estimate (INEGI)	monthly
Financial variables		
IPC	Stock exchange index	annual
E	Exchange rate	monthly
IR 28	Short-term interest rate	annual
SP 500	S&P 500 Index	annual
Cards	Credit/debit card transactions	monthly
SPEI	Interbank electronic transfers	monthly

Trans.: transformation applied to maximise correlation with IGAE growth. All series seasonally adjusted via X-13ARIMA-SEATS.

MACROECONOMIC VARIABLES PANEL



FINANCIAL VARIABLES PANEL



DFM extraction

DFM

- Formally, let $\mathbf{X}_t$ be the set of data, estimate the DFM:

$$\mathbf{X}_t = \mathbf{PF}_t + \varepsilon_t$$

- Idiosyncratic component: $e_{it} = x_{it} - F_t' \lambda_i$

DETERMINING THE NUMBER OF FACTORS BAI & NG (2002) IC

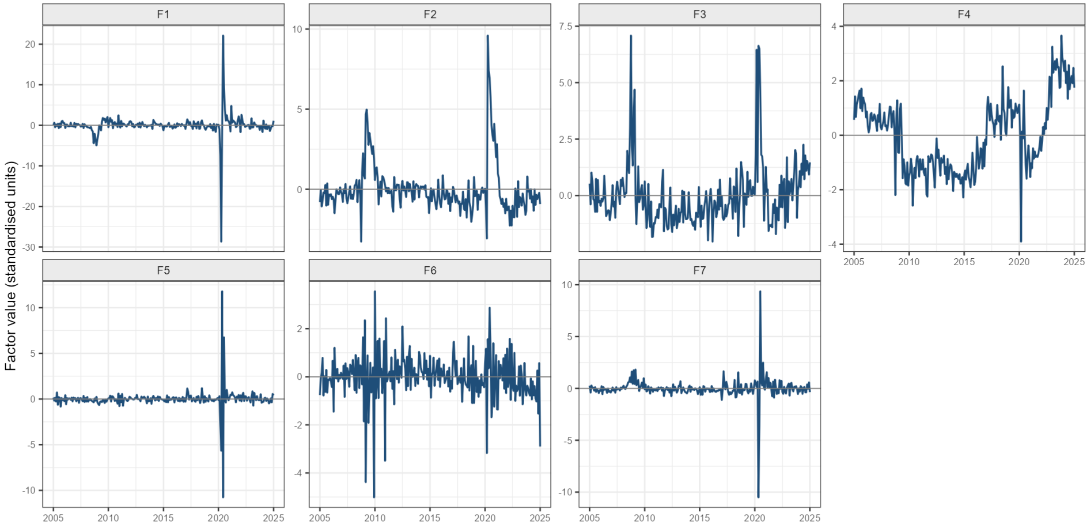
r	**IC1**	**IC2**	**IC3**
0	−0.004	−0.004	−0.004
1	−0.191	−0.186	−0.200
2	−0.200	−0.192	−0.219
3	−0.205	−0.192	−0.233
4	−0.208	−0.190	−0.245
5	−0.215	−0.193	−0.262
6	−0.218	−0.192	−0.274
7	−**0.233**	−**0.202**	−0.298
8	−0.228	−0.193	−**0.302**

IC1 and IC2 select $r = 7$. IC3 suggests $r = 8$.

- Seven factors explain **66.9%** of total panel variance.
- The remaining **33.1%** is idiosyncratic variance — to be used for the GARCH.

DFM Common Factors — Time Series

7 factors



Uncertainty extraction

UNCERTAINTY EXTRACTION — GARCH(1,1) ON IDIOSYNCRATIC COMPONENTS

Step 1 — Idiosyncratic residual

After the DFM, the idiosyncratic component for variable *i* at time *t* is: $e_{it} = x_{it} - \mathbf{F}_t' \lambda_i \, e_{it}$.

Step 2 — GARCH(1,1) per variable

For each variable *i*, fit: $e_{it} = \sigma_{it} \eta_{it}, \qquad \eta_{it} \sim iid(0, 1)$

$$\sigma_{it}^2 = \omega_i + \alpha_i \, e_{i,t-1}^2 + \beta_i \, \sigma_{i,t-1}^2$$

σ_{it} is the **conditional standard deviation** of the unforeseeable component (JLN measure of uncertainty).

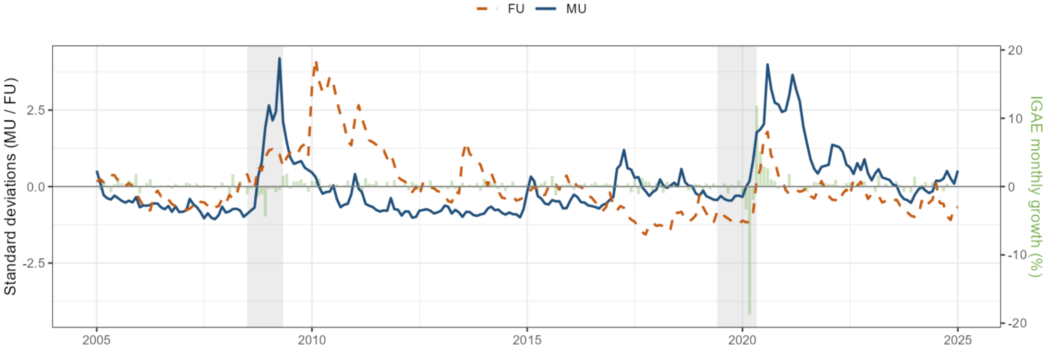
Step 3 — Aggregate into MU and FU

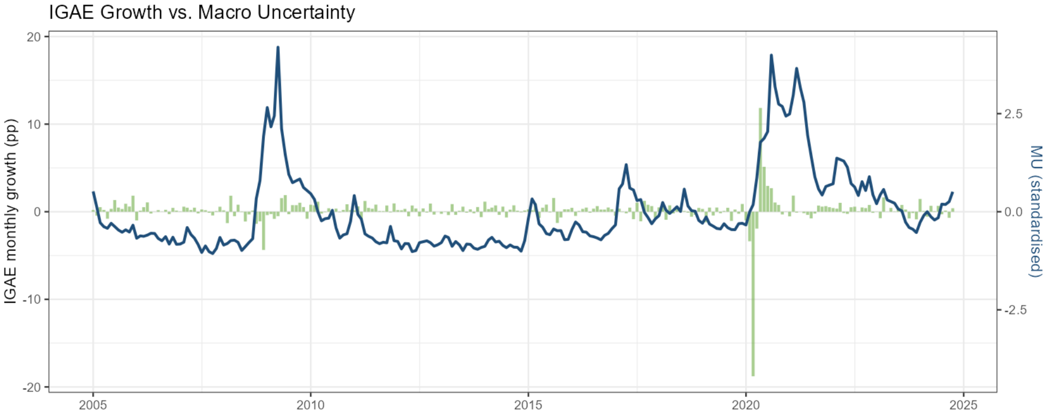
$MU_t = \frac{1}{N_M} \sum_{i \in M} \sigma_{it}, \quad FU_t = \frac{1}{N_F} \sum_{i \in F} \sigma_{it}$

Both indices are standardised to zero mean and unit variance.

Macro and Financial Uncertainty Indices

Bars: IGAE monthly growth | Lines: MU (solid) and FU (dashed)




IGAE Growth vs. Macro Uncertainty
IGAE monthly growth (pp)
MU (standardised)
20
10
0
-10
-20
2.5
0.0
-2.5
2005
2010
2015
2020
2025

SVAR

VAR LAG SELECTION CRITERIA

Model

$$\mathbf{X}_t = \begin{pmatrix} MU_t \\ IGAE_t \\ FU_t \end{pmatrix},$$

$$\mathbf{X}_t = \mathbf{c} + \sum_{j=1}^{p} \mathbf{A}_j \mathbf{X}_{t-j} + \mathbf{u}_t$$

Information criteria ($p_{\max} = 6$)

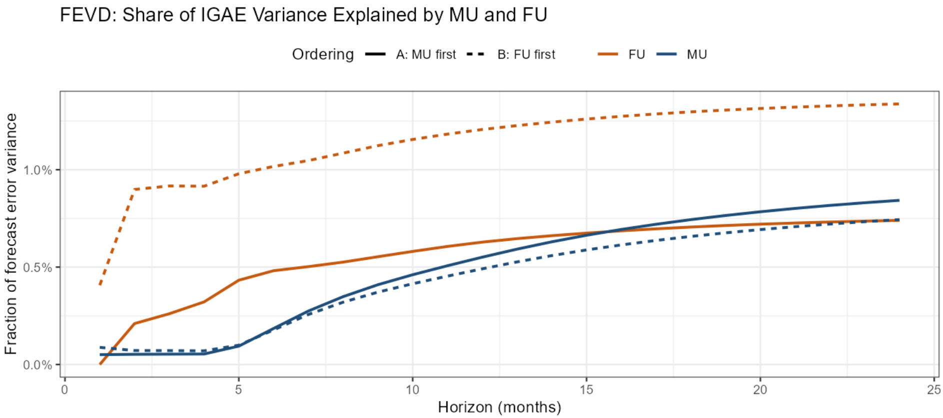
p	**AIC**	**HQ**	**SC**
1	−3.42	−3.35	**−3.24**
2	−3.48	**−3.36**	−3.17
3	−3.44	−3.26	−2.99
4	**−3.51**	−3.28	−2.93
5	−3.50	−3.21	−2.78
6	−3.47	−3.12	−2.62

- SC selects $p = 1$; HQ selects $p = 2$; AIC selects $p = 4$.
- With $p = 1$ and $K = 3$ variables the VAR has $K \times (K + 1) = 12$ free parameters.
- Choose 1 lag.

IRF

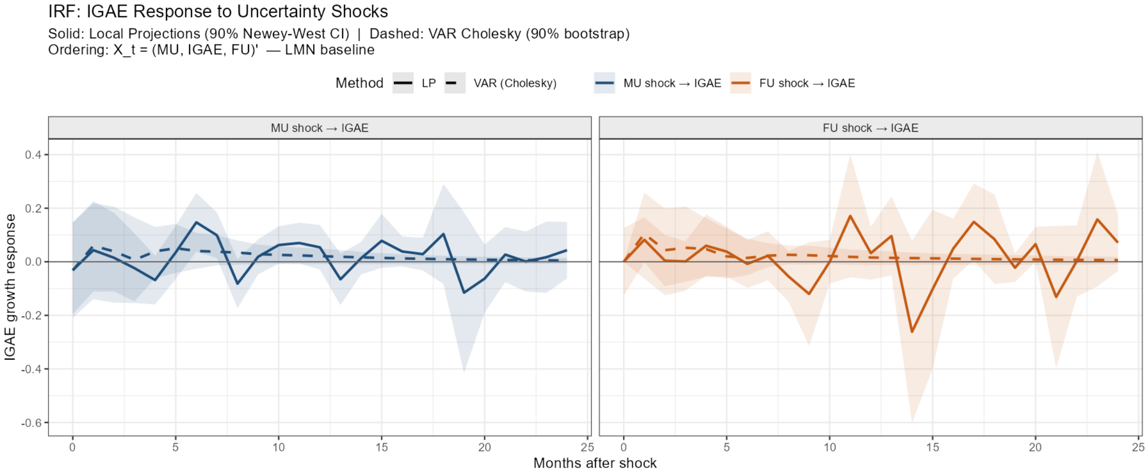
IRF: Response of IGAE Growth to Uncertainty Shocks
90% bootstrap confidence bands | 500 replications
MU shock → IGAE
FU shock → IGAE
A: MU first (LMN baseline)
B: FU first (alternative)
IGAE response
0.3
0.2
0.1
0.0
-0.1
-0.2
0
5
10
15
20
25
Months after shock

FEVD



FEVD: Share of IGAE Variance Explained by MU and FU
Ordering
A: MU first
B: FU first
FU
MU
Fraction of forecast error variance
1.0%
0.5%
0.0%
0
5
10
15
20
25
Horizon (months)

IRF WITH LP


IRF: IGAE Response to Uncertainty Shocks
Solid: Local Projections (90% Newey-West CI) | Dashed: VAR Cholesky (90% bootstrap)
Ordering: X_t = (MU, IGAE, FU)' — LMN baseline
Method
LP
VAR (Cholesky)
MU shock → IGAE
FU shock → IGAE
MU shock → IGAE
FU shock → IGAE
IGAE growth response
0.4
0.2
0.0
-0.2
-0.4
-0.6
0
5
10
15
20
25
Months after shock

Conclusions

MAIN TAKEAWAYS

- DFM is a powerful tool applicable for many purposes: index construction, nowcasting, uncertainty.
- The methodology of JLN/LMN is useful for measurement of unobserved variables: uncertainty.
- SVAR provides a framework to identify causal relations in these variables.

NEXT STEPS

- Perform the GaR methodology (Adrian, et al. 2019)
- Perform Cyclical + Permanent decomposition
- Elaborate cointegration analysis with the variables in levels
- Compare the results with those from Salgado and Trujillo (2025)

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